

FDS EXPERIMENT-12

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Experiment to detect outliers in given dataset.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Marks" : np.array([45,50,52,48,49,51,47,53.50,48,52,49,150])
}

df = pd.DataFrame(data)
print(df)
```

...	Marks
0	45.0
1	50.0
2	52.0
3	48.0
4	49.0
5	51.0
6	47.0
7	53.5
8	48.0
9	52.0
10	49.0
11	150.0

```
Q1 = df["Marks"].quantile(0.25)
Q3 = df["Marks"].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[
    (df["Marks"] < lower_bound) |
    (df["Marks"] > upper_bound)
]
```

```
▶ print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)
print("Outliers:")
print(outliers)
```

```
... Q1: 48.0
    Q3: 52.0
    IQR: 4.0
    Lower Bound: 42.0
    Upper Bound: 58.0
    Outliers:
         Marks
11  150.0
```

```
plt.boxplot(df["Marks"])  
plt.title("Box Plot for Outlier Detection")  
plt.ylabel("Marks")  
plt.grid(True)  
plt.show()
```

...

